

VIJAYASURIYA SURESH

JERSEY CITY, NJ

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Professional Summary

Dynamic AI and Data Science professional with expertise in neural network development and AI applications within the automotive and technology sectors. I am skilled in using machine learning for predictive maintenance and anomaly detection, proficient with AWS for scalable AI deployments, and adept in enhancing decision-making through NLP, OpenCV and data visualization. Proven leader in driving projects that improve operational efficiencies and customer satisfaction.

SKILLS

Languages: Python, SQL, JavaScript, HTML, XML, No-SQL, Java, C, C++.

Programming Paradigms: Object-Oriented Programming, Dynamic Programming.

Data Visualization & BI Tools: Power BI, Tableau.

Cloud & Data Architecture: AWS (S3, RDS, Glue, Databricks, Redshift, Bedrock, SageMaker, CloudTrail, Lambda)

Version Control: Git commands.

Data Science & Machine Learning: NumPy, Pandas, Matplotlib, Seaborn, Scikit Learn, OpenCV, OCR, Plotly, Regression Models, Decision Trees, Random Forest, K-means Clustering, Gradient Boosting, Neural Networks, RNN, CNN, NLP, LLMs.

Data & ML Pipelines: ETL Data Pipeline, CI/CD Data Pipelines, Docker, Kubernetes, Model Generation and Deployment.

Testing & Analysis: A/B Testing, Unit Testing, Ad-Hoc Analysis.

APIs: RESTful API, Graph API, Zen Desk API, Amazon S3 API, AWS SDK API, AWS API Gateway, AWS Lambda API, AWS Glue API, AWS Databricks API, AWS CloudTrail API.

Project Management: JIRA, Blue Ocean Strategy, SCRUM, Agile, Waterfall, Product Life-Cycle Management.

Performance: Defining KPI’s, Defining OKR’s, SWOT Analysis.

Soft Skills: Critical Thinking, Problem Solving, Strategic and Inquisitive mind, Quantitative and Qualitative Skills, Time Studies Performance, Adaptability & Flexibility, Leadership & Team Building, Communication & Interpersonal Skills, Decision-Making & Prioritization, Conflict Resolution, Time Management, Attention to Detail, Emotional Intelligence.

EXPERIENCE

Deloitte Consulting LLP, Tampa, FL June 2024 - Present
Machine Learning Engineer, Sr. Associate

Description: Deloitte Consulting LLP is a global consulting firm that offers a wide range of services to organizations across various industries. It provides strategic, technology, human capital, and operational consulting services, helping clients solve complex business challenges, improve performance, and drive innovation. Deloitte Consulting assists businesses with digital transformation, data analytics, cloud computing, cybersecurity, and supply chain management, among other areas. As part of Deloitte Touche Tohmatsu Limited (Deloitte), it is known for combining deep industry expertise with cutting-edge technology solutions to deliver measurable results.

Responsibilities:

- Designed and deployed NLP-driven models for automating client documentation classification across industries using AWS Bedrock and SageMaker. Leveraged BERT and GPT-3 for advanced text processing, achieving a 30% improvement in classification accuracy and processing efficiency for client deliverables.
- Engineered predictive maintenance solutions for IoT-enabled systems in client industries, utilizing AWS Bedrock, SageMaker, and Lambda. Applied NLP techniques and tools like Pandas and NumPy to streamline data transformations, reducing operational downtime by 40%.
- Developed a real-time threat detection system for client networks, integrating AWS SageMaker, CloudTrail, and Lambda. Enhanced security measures through GPT-3-based NLP embeddings, improving detection accuracy by 25% and ensuring timely threat alerts.
- Optimized image classification pipelines for client solutions using OpenCV and AWS SageMaker, integrating LLM-based embeddings to achieve a 35% improvement in accuracy and enabled real-time reporting via AWS Lambda.
- Implemented multi-cloud architectures for deploying AI models at scale, utilizing OpenCV for OCR and object tracking, automated with AWS API Gateway and CloudTrail to enhance data processing and increase system uptime by 50%.
- Developed CI/CD pipelines for MLOps workflows tailored to computer vision tasks, using Git for version control and AWS Lambda for automation, with TensorFlow and OpenCV to efficiently deploy optimized vision models into client systems.

Vyopta Incorporated, Austin, TX January 2024 - May 2024
Data and AI Engineer

Description: Vyopta Incorporated is a technology company that specializes in providing solutions for monitoring, analytics, and optimization of unified communication and collaboration systems. Its platform helps businesses track and manage the performance of video conferencing, VoIP, and other communication tools in real-time. Vyopta's software provides insights into system usage, network performance, and user experience, allowing organizations to improve efficiency, reduce downtime, and ensure seamless communication across teams. The company serves industries such as healthcare, education, and enterprise sectors, offering tools that enhance collaboration and support digital transformation initiatives.

Responsibilities:

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- Developed and deployed computer vision models utilizing TensorFlow for object tracking and OCR capabilities to automate the extraction of visual data from documents and objects during migration processes, leading to a 40% boost in operational efficiency by reducing manual data entry and speeding up data throughput.
- Engineered advanced computer vision models for trend analysis and feature extraction, using TensorFlow and PyTorch to analyze visual data trends and extract meaningful features. These models were integrated with AWS Databricks and AWS Glue to handle and analyze large-scale data sets efficiently, significantly enhancing operational efficiency and generating valuable insights from visual data patterns.
- Enhanced go-to-market strategies by developing real-time computer vision models using GPT-3 and RAG pipelines, leveraging AWS API Gateway for secure and efficient data exchanges, achieving a 95% increase in responsiveness and enabling AI-driven business intelligence.
- Created a centralized, intelligent documentation system utilizing large language models and computer vision techniques with AWS SageMaker and RDS, implementing GPT-3 embeddings for semantic search and integrating AWS CloudTrail for logging and audit tracking, optimizing data retrieval and storage processes.

Toyota Finance services, Plano, Texas

December 2022 – December 2023

Data Scientist

Description: Toyota Financial Services (TFS) is the financial services arm of Toyota, offering a wide range of financing solutions to customers and dealerships. TFS provides vehicle leasing, retail loans, and dealer financing, enabling customers to purchase or lease Toyota and Lexus vehicles. The company also offers insurance products and extended service plans. Through its services, Toyota Financial Services helps make vehicle ownership more accessible, supporting both individual consumers and businesses in managing their automotive financing needs. TFS operates globally, with a focus on enhancing the customer experience and promoting Toyota's commitment to mobility

Responsibilities:

- Developed predictive maintenance systems using AWS Bedrock, SageMaker, and Lambda for financial asset monitoring. Applied NLP techniques such as tokenization and BERT embeddings, enabling real-time equipment analysis and reducing operational downtime by 40%.
- Automated financial forecasting models to analyze large volumes of textual and numerical financial data. Leveraged Python and LLMs (e.g., GPT-3) for data transformation, feature engineering, and forecasting, improving accuracy and processing speed for strategic financial planning.
- Built a real-time speech-to-text application to streamline loan customer interactions, using Python, AWS API Gateway, and NLP models. Enhanced transcription accuracy by integrating BERT embeddings and Llama Index, enabling robust, real-time performance for voice-driven loan services.
- Designed a sentiment analysis tool for financial customer service. Leveraged BERT-based sentiment models with AWS SageMaker, AWS Lambda, and Power BI dashboards to monitor customer feedback in real-time, improving response times and increasing customer satisfaction.
- Developed a multi-language chatbot for customer support using AWS SageMaker and Lambda, integrating OpenCV for text recognition to automate the resolution of financial service queries, achieving a 30% reduction in response time by automating text extraction and analysis.
- Engineered an automated ETL pipeline for real-time processing of financial data streams using AWS Glue and Databricks, with CI/CD pipelines and incorporated OpenCV for data validation to enhance data accuracy, reducing processing time by 40% and improving system scalability.
- Implemented a fraud detection framework using AWS Bedrock, SageMaker, and XGBoost, utilizing OCR and object tracking via OpenCV to analyze transactional data for anomalies, which increased fraud detection accuracy by 25% and enabled real-time reporting with Power BI dashboards.
- Integrated a cloud-based inventory and loan management system using AWS API Gateway and Redshift, with TensorFlow and Random Forest models enhanced by OpenCV for computer vision-driven demand forecasting, resulting in a 35% increase in resource allocation accuracy.

Amazon, Chennai, India

July 2020 - July 2022

Data Scientist

Description: Amazon is a global technology and e-commerce company that offers a wide range of products and services. It operates the world's largest online marketplace, where customers can purchase everything from books and electronics to clothing and groceries. In addition to e-commerce, Amazon provides cloud computing services through Amazon Web Services (AWS), which is a leader in the industry. The company also produces consumer electronics like the Amazon Echo, develops digital streaming services through Amazon Prime Video, and offers logistics, artificial intelligence, and entertainment solutions. Amazon is known for its innovation, customer-centric approach, and diverse business ventures

Responsibilities:

- Developed a personalized recommendation system using clustering analysis and behavioural insights. Enhanced user engagement by 70% through advanced data visualization techniques, leveraging Python for data processing and AWS API Gateway for seamless data interchange.
- Optimized dynamic pricing models through trend analysis and statistical models using Python, which led to a 50% increase in conversion rates. These models were deployed on AWS Lambda, interacting seamlessly with AWS S3 and

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Redshift for robust data storage and analysis.

- Conducted detailed logistics analysis using time-series data to enhance warehouse throughput by 40%, employing Python with AWS Glue for effective data integration and AWS Databricks for efficient processing, ensuring streamlined data flow and thorough analysis.
- Implemented real-time security monitoring systems using Power BI and Python scripts, interfaced with AWS API Gateway. This setup significantly enhanced multi-factor authentication processes by providing dynamic, real-time security dashboards.
- Integrated augmented reality insights into shopping experiences by processing AR data through Python scripts stored in AWS S3, displayed via Power BI dashboards, effectively boosting user engagement by 15%.Analysed IoT device data for smarter device management, employing Python for statistical analysis and AWS Lambda for real-time data processing, demonstrating effective use of AWS CloudTrail for logging and monitoring device activity.
- Enhanced image classification systems using Support Vector Machines and Random Forest algorithms in Python, with model training and deployment on AWS using RDS for data management and Lambda for processing, improving classification accuracy by 35%.
- Deployed a multi-cloud architecture to facilitate ML model deployment using logistic regression and decision trees, ensuring a 60% increase in system uptime by leveraging AWS services such as S3, Lambda, and CloudTrail for robust, scalable infrastructure management.

EDUCATION

Master of Science in Data Analytics and Visualization

Yeshiva University – Katz School of Science and Health, New York, NY

Bachelor of Engineering in Computer Science and Engineering

Anna University - Velammal College of Engineering and Technology, Madurai, TN, India

PROJECTS

- **NYC Weather Data Analysis:** Developed a project on weather forecasting through advanced data analysis and visualizations of historical weather patterns with its change according to global warming over time, enabling more precise predictions.
- **Hospital System Automation:** Spearheaded a project on automated integration of AI in hospital operations, increasing efficiency by 30% with real-time data analytics, enhancing patient care and resource allocation.
- **Fraud Detection System:** Crafted a project on high-accuracy fraud detection system using neural networks to analyse transactional data, significantly reducing financial risks by identifying anomalous patterns.
- **Netflix Recommendation System:** Developed a Netflix's recommendation engine by analysing viewer preferences and engagement data on kagle for improved recommendation accuracy enhancing user experience in content selection.
- **Airline Fare Prediction Model:** Built a predictive model for airline fares using historical pricing and market trends analysis, helping airlines strategically adjust fares based on demand and competition.
- **Amazon Prime Video Insights:** Developed a comprehensive data model to decipher viewing habits on Amazon Prime, influencing content strategy and marketing through detailed user engagement analysis.
- **Automated Sentiment Analysis for Marketing:** Implemented an NLP-driven tool to analyse customer feedback across platforms, enabling targeted marketing strategies and product adjustments based on customer sentiment.
- **Golden Craft Marketplace:** Delivered strategic analytics for a novel gold jewellery marketplace, driving customer engagement and acquisition by aligning product offerings with unique market demands through data-driven insights.

Certifications: Google Data Analytics, AWS Cloud Practitioner, Google Advanced Data Analytics, Meta Database Engineer, Google Business Intelligence, IBM Product Management, Google Project Management.